

INDIAN STATISTICAL INSTITUTE

End Semester : Second Semester 2024-25

Course : M.Math First Year

Subject : Functional Analysis

Instructor : Partha Sarathi Chakraborty

Date : 28st April 2025

Marks :50

Duration: 180 minutes

1. There are two sections. You can answer as much as you wish from section A but maximum you can score from this section is 35.
2. Notation: Throughout this paper $\mathcal{H}$ stands for a Hilbert space and $\mathcal{B}(\mathcal{H})$ stands for bounded linear operators on $\mathcal{H}$.

Section: A

1. Let $T \in \mathcal{B}(\mathcal{H})$ be normal, i.e., $TT^* = T^*T$. Show that

(a) $\ker(T) = \text{Ran}(T^*)^\perp, \ker(T^*)^\perp = \overline{\text{Ran}(T)}$.

(b) $\ker(T) = \ker(T^*)$.

(c) $\sigma_r(T) = \emptyset$, where $\sigma_r(T)$ is the residual spectrum of T . [4+4+4=12]

2. Let $\mathcal{H}, \mathcal{K}$ be Hilbert spaces and $U : \mathcal{H} \rightarrow \mathcal{K}$ be a bounded linear map. Then show that the following are equivalent.

(a) $UU^*U = U$.

(b) $U^*UU^* = U^*$.

(c) UU^* is an orthogonal projection.

(d) U^*U is an orthogonal projection. [12]

3. Let E be a $\mathbb{C}$ vector space and $\mathcal{A}$ be a subspace of the space of linear maps from E to $\mathbb{C}$. Show that $(E, \sigma(E; \mathcal{A}))^* = \mathcal{A}$. [8]

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$x, y \in \mathcal{H}$
 $\langle Tx, y \rangle = \langle x, T^*y \rangle$
 $\langle T^*y, x \rangle = \langle y, T x \rangle$
 $T^*: \mathcal{H} \rightarrow \mathcal{H}$
 $f \mapsto f \circ T$

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4. Let E, F be Banach spaces and $T : (F^*; \text{weak}^*) \rightarrow (E; \text{weak}^*)$ be a continuous linear map. Show that there exists a norm continuous linear map $S : E \rightarrow F$ so that $T = S^*$. [8]

5. Let $\{\lambda_n\}$ be a sequence of scalars so that $|\lambda_n| \downarrow 0$. Define $T : \ell_2(\mathbb{N}) \rightarrow \ell_2(\mathbb{N})$ to be the linear map

$$T(\{x_n\}_{n \in \mathbb{N}}) := \{y_n\}_{n \in \mathbb{N}}, \text{ where } y_n = \lambda_n x_{n+1}, \forall n \in \mathbb{N}.$$

Compute $\|T^n\|$ and conclude that $\lim_n \|T^n\|^{1/n} = 0$. Such operators are called quasi nilpotent operator. Let $\mathcal{A} \subseteq \mathcal{B}(\ell_2(\mathbb{N}))$ be the unital Banach algebra generated by T and $\omega : \mathcal{A} \rightarrow \mathbb{C}$ be a unital homomorphism. Show that $\omega(T) = 0$. [5+5=10]

Section: B

6. Let $\mathcal{H}_1, \mathcal{H}_2 \subseteq \mathcal{H}$ be closed subspaces of the Hilbert space $\mathcal{H}$. Suppose $\dim \mathcal{H}_1 < \infty$ and $\dim \mathcal{H}_1 < \dim \mathcal{H}_2$. Then show that $\mathcal{H}_1^\perp \cap \mathcal{H}_2 \neq \{0\}$. [5]
7. Let $T \in \mathcal{B}(\mathcal{H})$ be a normal operator with associated unique regular spectral measure $\xi : \mathcal{B}_{\sigma(T)} \rightarrow \mathcal{P}(\mathcal{H})$, where $\mathcal{B}_{\sigma(T)}$ is the collection of all Borel subsets of $\sigma(T)$ and $\mathcal{P}(\mathcal{H})$ is the collection of orthogonal projections on $\mathcal{H}$. Let $A \in \mathcal{B}_{\sigma(T)}$ be a non-empty subset.
- (a) Show that $T(\xi(A)(\mathcal{H})) \subseteq \xi(A)(\mathcal{H})$.
- (b) Let $S = T|_{\xi(A)(\mathcal{H})} : \xi(A)(\mathcal{H}) \rightarrow \xi(A)(\mathcal{H})$. Show that $\sigma(S) \subseteq \overline{A}$. [2+8=10]
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☺ Good Luck ☺